LUCILA VÁZQUEZ SORIANO

Junior Data Scientist · NLP & Deep Learning · Alicante, Spain · lucilavsoriano@gmail.com · LinkedIn · Portfolio

Profile

Data scientist with an MSc in Data Science (University of Alicante, 2026) and an engineering degree. Built end-to-end deep learning projects in NLP, computer vision and causal inference. Strong probability and statistics foundation: calibration, Bayesian methods and causal estimation. AWS-certified in Generative AI Foundations, including prompt engineering. Eager to take NLP and LLM solutions from research to production. Both theses graded 10/10; co-author at SOCO 2025 (Springer). English C1.

Education

- **MSc in Data Science**, University of Alicante · 2025 – 2026 · *Thesis: deep learning for gastric cancer histopathology (10/10)*
- **BSc in Multimedia Engineering**, University of Alicante · 2019 – 2025 · *Thesis: VR escape rooms to prevent cognitive decline (10/10)*

Data & AI Projects

SemEval 2023 Task 10 — Explainable online sexism detection (NLP) · *Hugging Face Transformers, scikit-learn, Gradio*

- Achieved 0.80 macro-F1 on binary sexism detection (DistilBERT) and 0.44 on 4 categories (hierarchical RoBERTa).
- Beat the best TF-IDF baseline (SVC + oversampling, 0.73) by 7.5 pp after testing class-balancing strategies.
- Built a Gradio demo to test the model interactively on new text.

MSc Thesis — Gastric cancer classification in histopathology · *PyTorch (timm), CatBoost, SHAP, Docker*

- Reached 94.70% accuracy (+1.5 pp over baseline) by adding a CBAM attention module to EfficientNetV2-B0.
- Ensured robust results on 245,196 patches with stratified 5-fold CV, 5 seeds and duplicate detection.
- Gained ~1 pp accuracy at every patch size by applying data augmentation.
- Made predictions interpretable with SHAP and attention maps, documenting model limitations.

Causal inference: built environment and cycling in London · *Causal Forest, PSM, g-computation, semantic segmentation*

- Quantified heterogeneous effects (CATE) with Causal Forest and PSM: bike lanes raise cycling, steep slopes cut it.
- Built a 2008–2024 dataset combining official counts, Street View imagery and OpenStreetMap.

Model calibration and graph neural networks · *TensorFlow/Keras, PyTorch Geometric*

- Reduced test ECE by 86% on Fashion-MNIST with Focal Loss + Temperature Scaling, also comparing MC Dropout.
- Raised MUTAG accuracy from 74.0% to 79.4% by adding Laplacian positional encodings to GCNs (5 benchmarks).

Serverless people-detection pipeline on Azure (production) · *Azure Functions, Blob Storage, Azure AI Vision*

- Deployed an event-driven pipeline that captures a frame every 0.5 s and processes it with Blob-triggered Azure Functions.
- Automated publication of detection results (images and CSV) to a static web app.

Urban and geospatial analytics (Torrevieja, Madrid) · *GeoPandas, OSMnx, Plotly*

- Estimated effective population and tourist-housing pressure by integrating FRONTUR, INE and census data.
- Co-built a “15-minute city” accessibility index for Madrid’s neighbourhoods in a 4-person team.

Experience

Software Developer Intern, Doña Rosa S.L. · Oct – Dec 2023

- Enabled real-time client-server communication in Unreal Engine by integrating Socket.IO with web servers in C++.
- Migrated projects across Unreal Engine versions and tested web applications.

Student Guide and Representative, University of Alicante · Dec 2022 – Apr 2023

Skills

- **Languages:** Python, SQL, R, C++, Java, JavaScript
- **NLP & GenAI:** Hugging Face Transformers, Sentence Transformers embeddings, TF-IDF, prompt engineering
- **Deep learning:** PyTorch, TensorFlow/Keras, PyTorch Geometric, CNNs, transfer learning, attention mechanisms
- **Probability & statistics:** Bayesian methods, count models, causal inference (PSM, Causal Forest), model calibration
- **Production & cloud:** Docker, Git, Azure Functions, AWS, Gradio, ETL pipelines, APIs
- **Collaboration:** Scrum, Kanban, team projects

Publication, Certification and Languages

- *VR Escape Room to Prevent Cognitive Decline in Older Adults* — SOCO 2025, Springer (co-author)
- Generative AI Foundations, AWS (IMMUNE): prompt engineering, responsible AI, bias and security
- Spanish (native) · English (C1) · Catalan (C1) · Japanese (A2)